

Real Atoms

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Real Atoms

Unfortunately, real atoms (including the hydrogen atom) are not very well described by the simple theory. The following contribute to actual structure:

Higher Order Effects

Multiple electrons

Spin-Orbit coupling

Relativistic

Nuclear spin

Zeman effect

Quantum mechanics of identical particles

Pauli exclusion principle

Spin-Orbit Coupling

- Electrons have spin, and thus angular momentum, as does electrons orbiting the nuclei. These two motions generate magnetic fields that interact.
- The total angular momentum vector is

$$\vec{J} = \vec{L} + \vec{S}$$

J is Quantized

- One can show that $J = \sqrt{j(j+1)}\hbar$
 $J_z = m_j\hbar$

- and $m_j = -j, -j+1, \dots, j-1, j$

$$l = 0 \qquad j = 1/2$$

$$l \neq 0 \qquad j = l \pm 1/2$$

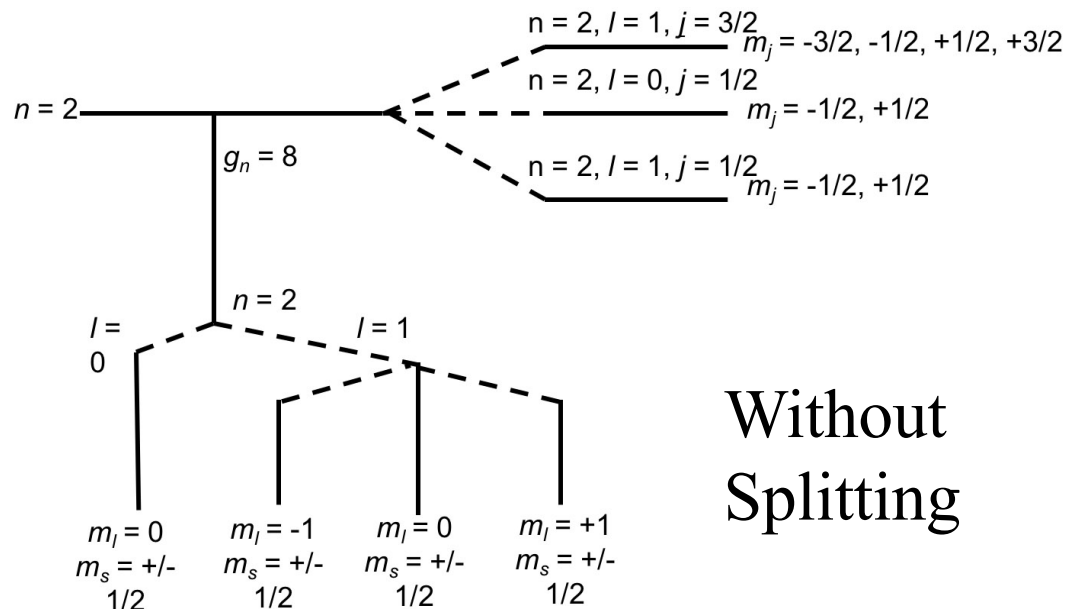
Good Quantum Numbers

- Thus m_l and m_s cease to be “good” quantum numbers and are replaced by j and m_j .
- Good quantum numbers are thus

$$n, l, j, m_j$$

Splitting

- The Spin-orbit coupling causes quantum states that would otherwise have the same energy to “split”, thus changing the degeneracy as well.



With
Splitting

Without
Splitting

Sodium

